AI Detection Biases with the Rise of LLMs: Testing GPTZero as an AI Detection System on Two Sentence Horror Stories

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Abstract

The rise of large language models has introduced new challenges in distinguishing human-written content from AI-generated text, raising concerns about misinformation, authorship, and the ethical use of automated detection tools. This study investigates the effectiveness of GPTZero and ChatGPT 4.0 mini as AI detection systems when applied to short-form creative writing, specifically two-sentence horror stories collected from Reddit and generated through the ChatGPT API. A dataset of 900 stories was compiled across human-written posts from 2020 and 2025 and AI-generated samples, and each story was evaluated using GPTZero's perplexity and burstiness metrics as well as ChatGPT's binary classification. Our results show that GPTZero reliably identifies AI-generated stories but frequently misclassifies human-written stories as AI, producing a notably high false-positive rate. Conversely, ChatGPT overwhelmingly labels stories as human, failing to detect nearly all AI-generated examples. These findings highlight significant shortcomings in current detection systems, particularly for short and stylistically simple texts, and emphasize the need for more transparent, context-aware, and ethically aligned approaches to AI-generated content detection.

1 Introduction

Large language models (LLMs) have exploded since the initial creation of ChatGPT. Now with LLMs' pervasiveness, they generate anything from large research papers to small doodles used in a child's social media post. Thus, there is a general lack of credibility in what media is created by a human or an LLM. This problem is exacerbated in the academic context when credibility and plagiarism is important. Prior to LLMs, plagiarism detectors have been used to ensure students use their original work. In the present day, many companies and organizations have collaborated with schools to create AI detection systems to predict whether a given text is generated by AI or is made by humans.

These detections systems are already tested by the public to not be the most accurate, especially when users write in very generic foundations. Students argue that cover letters and five page essays become tagged as AI when they follow school system's templates. These false positive rates raise the question of whether it is ethical to use such detection systems. Because of the lack of accuracy in these systems, many schools have stopped using them. However, the general public still uses them and argues with the position of the detection system instead of delving deeper into the text itself.

In fact, these systems perform much worse with shorter lengths of text. Thus this paper delves into the performance of the AI detection system GTPZero to see how well a state-of-the-art system would perform with short fictional stories. Specifically two sentence horror stories from Reddit and generated by ChatGPT will be tested in GPTZero and ChatGPT. Two sentence horror stories are short fictional stories that base creative horror in two sentences. They have been popularized by the active subreddit TwoSentenceHorror as they have to be creative enough to spark a scary plot in only two sentences.

2 Related Work

Recent research in AI detection has focused on more long form texts. More specifically most academic papers consist of research on AI detection systems testing on high level essay papers. Thus there is a significant lack of research of AI detection systems on short form content especially in fiction. Fortunately, there is plenty of recent research on AI detection systems; thus, their methodologies can be translated to follow the same format to test how well these systems perform.

A leading paper in the educational area describes the testing of twelve public detection systems and two commercial systems. Of these systems includes GPTZero, which will be the main testing system for the Two Sentence Horror stories. The paper (Debora Weber-Wulff, 2023) has individual authors write their own stories and use ChatGPT to generate undergraduate level papers of over 10,000 characters.

There is only one academic paper published in the realm of Two Sentence Horror Stories by Adam Keppler: a professor at Stanford University. The paper - HorrifAI: Using AI to Generate Two-Sentence Horror (Keppler and Chen, 2019)- attempts to create an AI system that would generate two sentence horror stories. Unfortunately there is no open source system nor data published by Keppler; however, a group of students at Northeastern University have created an open source model and Kaggle set that completes a similar mission. This system (A Vo, 2023) works on data pulled using PullPush from January 2015- April 2023. The base model will take in a first sentence as build up then the language model will create a second sentence to follow up. The combination of input and output will thus create a Two Sentence Horror story.

This paper will build on these efforts by developing a framework to pull horror stories to see how well detection services perform. The main focus will be to show the possible bias of over-utilizing detections systems as they may provide misinformation especially in short form fiction.

3 Approach

This paper aims to show the possible ineffectiveness of GPTZero as an AI detection system. In order to test this system, plenty of data has to be created and collected. Additionally there has to be multiple baselines to ensure possible biases may not occur. The main system will follow the process shown in Figure 1: PullPush will scrape Reddit posts from 2025 and 2020, ChatGPT will create AI stories, all of this data will be tested using GPTZero and ChatGPT.

Starting with the data scraping, PullPush is a web service widely used by researches to find Reddit posts using multiple filters. Additionally, there exists an open source version of this scraping system so mass amounts of posts can be scraped. BAScraper (maxjo020418, 2023) bases itself on both PullPush and Arctic-Shift. For this paper, PullPush was utilized to scrape Reddit posts from the subreddit TwoSentenceHorror. These posts will represent the human-made set of data. As LLMs have been created post 2021, there is a chance of these Reddit posts being AI generated. TwoSentenceHorror already has rules to prevent plagiarism and AI posting, yet posts from 2025 and 2020 were scraped. The recent posts ensures creativity and recency while the earlier posts were held as a baseline as those posts will never have been exposed to an LLM.

More specifically, three sets json files consisting of 100 posts were created for each year. 100 posts were created with dates prior of 12/1/2025, 8/1/2025 and 6/1/2025 for more recent posts while 100 stories from dates prior to 12/31/2020, 9/1/2020, and 6/1/2020. Thus there exists 600 total posts of human made Reddit stories. 6 extra json files were created with the same exact stories but formatted simpler to be processed by GPT 4.0 mini.

To test the AI generated stories, the GPT 4.0 mini api was called to create Two Sentence Horror stories. Specifically the api was called to create a json file of 100 two sentence horror stories that are at least 100 characters long. This prompt was run 3 times with a temperature of 1 to create the most creative stories to prevent duplicability.

GPTZero is originally a web service with closed processing. However, a GitHub user has created an open source GPTZero (BurhanUlTayyab, 2023) model that works with almost exact similarity to the web service. This system works with a huggingface system based on GPT2 to detect the perplexity and burstiness of inputted text. The perplexity is how unexpected the next word is compared to the previous context. The burstiness is the variation in sentence structure and format. The stories were formatted to ensure that they were over 100 characters long prior to being run in GPTZero. Since AI detection systems are based on context, GPTZero requires each input to be a minimum 100 characters else they will be skipped.

The GPT 4.0 mini api is then called upon to detect whether the stories are AI generated or human made. Specifically the system was prompted to read through the json files and label each story with a temperature of 0. The low temperature was chosen to prevent any inherent biases that may consistently exist so that the model will be able to take variety with little structual bias.

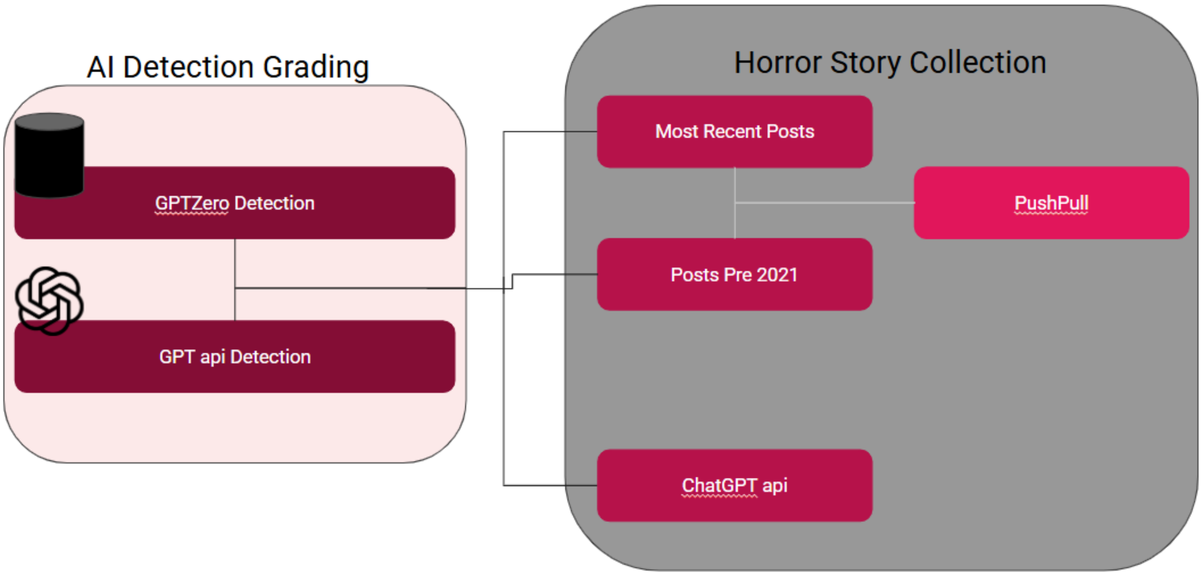

Figure 1: General process of collecting data from Reddit using PullPush api and generating stories from GPT 4.0 mini to grading their likeness using GPTZero and GPT 4.0 mini

4 Experiments

In short, the PushPull system will scrape 3 json files of 100 Two Sentence Horror stories from 2020 and 3 json files from of 100 from 2025. 3 separate json files from 2020 will consist simply of the 100 stories (title and selftext of the Reddit post) to be sent through GPT. The ChatGPT 4.0 mini api will run to create 3 json files consisting of 100 Two Sentence Horror stories. The original 2020, 2025, and ChatGPT json files will be sent straight to GPTZero to be tested. GPTZero will then create csv files of each json with 4 columns: the story, label(0 for AI and 1 for human), perplexity score, and burstiness score. The altered 2020 and 2025 json files along with the ChatGPT generated json files will be sent through the ChatGPT 4.0 mini system to be simply labeled in a csv with story and label (human or ai).

5 Analysis

As seen in Figure 2, GPTZero has a relatively good detection for AI generated stories. There is a lower level of detection of human made stories to be labeled as human than AI generated stories to be labeled as AI. Unfortunately, over half of the human made stories were detected to be AI generated. Overall, GPTZero performs relatively well at detecting when an AI generated story is AI, but performs quite poorly when testing to see if human made stories are tested for AI generation.

As seen in Figure 3, ChatGPT 4.0 mini works relatively well in determining when human made sto-

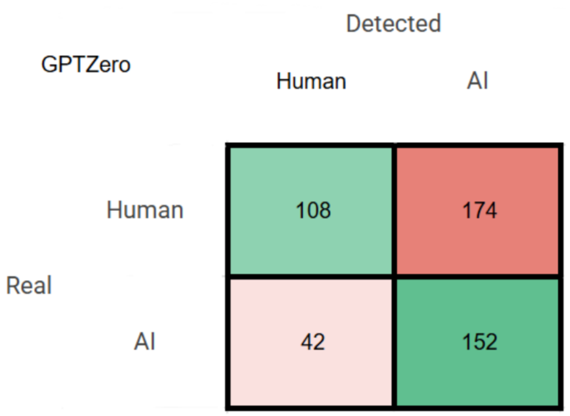

Figure 2: This figure represents the True Positives, False Positive, True Negatives, and False Negatives in a confusion matrix for the GPTZero detection. The actual human made (Reddit posts) are on the rows while the GPTZero detected values are on the columns.

ries are human made. However, the system seems to overfit this human detection as all but one story written by ChatGPT was detected as written by a human. This almost 100% chance of AI generated stories to be detected as human made is unfortunate for any user trying to find accurate results.

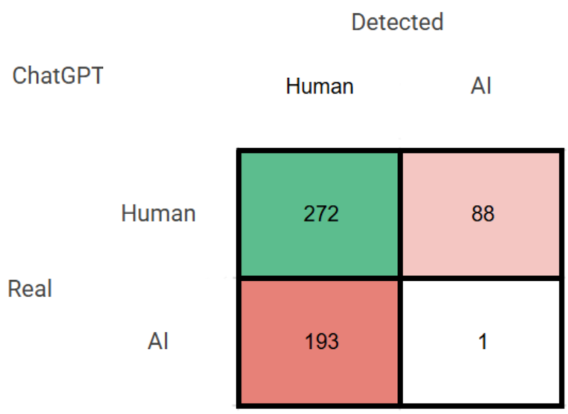

Figure 3: This figure represents the True Positives, False Positive, True Negatives, and False Negatives in a confusion matrix for the ChatGPT detection. The actual human made (Reddit posts) are on the rows while the GPT api detected values are on the columns.

6 Conclusion

All in all, AI detection systems do not perform as well as intended. Especially the LLM ChatGPT seems to overfit stories to mostly have stories be labeled as human made while GPTZero seems to mostly overfit to be labeled as AI generated.

6.1 Limitations

There were two main limitations in working on this research. The biggest one was that most systems cannot work with small amounts of text. Specifically for GPTZero it cannot run without 100 characters minimum. Thus many of the posts and ChatGPT generated stories were unusable. The second limitation was in finding the data that would also fit the criteria.

6.2 Future Work

This work can be expanded on to include other short systems including haiku poems. This project aims to open the AI detection systems to also work positively with short amounts of text.

Acknowledgments

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A Appendix